CHAPTER III

4.2(A) How do we prevent learning algorithms from acquiring morally objectionable biases?

For most of history, people have made decisions about things that affect other people. People made choices about things like hiring, getting loans, figuring out what diseases people had, retail, manufacturing, entertainment, and more. But in the last few decades, artificial intelligence (AI) has been able to do some things better and more reliably than people could. In 1997, for instance, Deep Blue, a computer chess program made by IBM, beat Garry Kasparov, the best chess player in the world. AI is now being used to make decisions in areas like hiring, loan eligibility, housing, medicine, technology, banking, marketing, and entertainment. AI is being used in these fields because it can do some tasks more accurately than people can. While AI algorithms can increase efficiency in hiring, entertainment, and other industries, these algorithms can also contribute to bias and discrimination, particularly when they are trained on biased data or lack proper oversight in their implementation, which can lead to unfair outcomes for certain groups of people. An algorithm is a step-by-step set of rules that a computer follows to learn patterns from data and make decisions or predictions. In AI systems, algorithms determine what the system learns, how it learns, and how it acts.

Historical records may exhibit previous prejudice, user behaviour might mirror societal preconceptions, and human-annotated information may encompass subjective evaluations or implicit biases. Bias is defined as "a predisposition of temperament or attitude" when training data exhibits social, cultural, or institutional biases. Consequently, preferential treatment of one entity over another constitutes bias. In reality, bias is intrinsically linked to discrimination, which is characterised as a "prejudiced or prejudicial perspective, conduct, or treatment." "Discrimination is a consequence of bias; a distorted perspective can result in discriminatory behaviour towards others.

AI systems acquire knowledge from sources including historical records, user behaviour, and human-annotated datasets. These data sources influence the manner in which algorithms identify patterns, generate predictions, and render choices, which can lead to biased outcomes if the underlying data reflects societal inequalities. Data, generated inside social, cultural, and institutional contexts, frequently mirrors prevailing human values, beliefs, and injustices; hence, AI systems tend to reproduce and exacerbate these biases instead of

rectifying them. As AI assimilates patterns from existing data, societal discriminatory practices—such as gender bias, racial prejudice, or economic inequality—are ingrained in algorithmic decision-making. Consequently, AI may produce inequitable results in domains such as recruitment, law enforcement, credit assessment, and healthcare, perpetuating existing disparities rather than fostering equity.

This bias may arise from conflating correlation with causation, being influenced by subjective personal characteristics, or relying on biased or insufficient data sets or human sources. The endeavour to address AI bias is continuous, and there are no straightforward remedies in this domain. AI prejudice may potentially face legal challenges, as discrimination against specific demographics might be subtly introduced. As society increasingly employs AI in diverse decision-making processes, the concern of bias must not be disregarded.

Artificial Intelligence (AI) systems discern patterns from data. Nonetheless, if the data embody prevailing social inequities or biases, the algorithm may assimilate and exacerbate them, leading to ethically problematic prejudices—such as racial profiling, gender stereotyping, or inequitable employment practices. Racial profiling denotes the prejudicial practice of singling out individuals for suspicion of criminal activity or examination predominantly based on their race, ethnicity, or perceived racial traits, rather than on specific facts. Consequently, mitigating such biases is both a technological and ethical problem.

An exemplary instance of discrimination is racism, wherein an individual of a specific race interacts differently with members of other races compared to her interactions with members of her race. Consequently, if she encounters an individual of a different ethnicity, she responds with disdain, hostility, and even aggression. It is a compelling inquiry to see if racism may exist as an attitude that remains unexpressed in action. I assert that this question can be answered affirmatively, indicating that racism does not inevitably result in an action. We can therefore differentiate between dispositional racism—the inclination to treat individuals of a different race differently than one treats those of one's own race—and racism in practice.

When a decision-making algorithm considers extraneous factors, bias induces discrimination against specific groups. The inclusion of pertinent parameters may enhance accuracy, whereas irrelevant parameters can detrimentally affect accuracy and exacerbate "racism, sexism, and other injustices" by leading to skewed outcomes that unfairly disadvantage marginalised groups. Irrelevant factors encompass those that exert no influence on the ultimate objective. Personal factors, such as nationality and race, do not influence an individual's skills. Supplying this knowledge to an algorithm will not enhance its accuracy, and the potential for prejudice or

discrimination renders these elements detrimental to AI decision-making, as they can lead to biased outcomes that unfairly disadvantage certain groups of people, such as minorities or under-represented communities, by perpetuating existing inequalities.

Ethnicity refers to the shared cultural, linguistic, ancestral, or social characteristics of a group. Ethnic bias in AI systems occurs when algorithms unfairly disadvantage or inaccurately portray individuals from particular ethnic groups.

How ethnicity-related bias arises in AI:

Biased training data: Historical data may reflect discrimination or under-representation of certain ethnic groups.

Sampling bias: Databases (such as facial recognition datasets) may not adequately represent some ethnic groups.

Design choices: Features or labels may implicitly encode ethnic markers, leading to unequal outcomes.

impacts:

Inaccurate facial recognition for certain ethnicities

Discriminatory outcomes in hiring, credit scoring, policing, or healthcare

Loss of trust and reinforcement of social inequalities

Face recognition algorithms exhibit elevated error rates for individuals from specific ethnic origins due to imbalanced training data, raising concerns about equity and civil liberties.

An AI model for decision-making may identify correlations that inadvertently arise among disparate groups, thereby intensifying discrimination against these groups, such as racial or gender biases that can lead to unfair treatment in areas like hiring or law enforcement. Consequently, eliminating personal characteristics from AI models helps mitigate discrimination against specific populations. Consequently, eliminating personal characteristics from AI models helps mitigate discrimination against specific populations. A pertinent source of bias in automated decision-making arises from distorted data sets that fail to appropriately reflect the target population of an algorithm. Consequently, algorithms may exhibit a higher incidence of errors concerning under-represented demographics within the data set, leading to unfair treatment and potentially harmful outcomes for those groups, such as increased rates of false negatives in predictive policing or biased loan approval processes.

AI biases may originate from the prejudices of humans involved in developing the algorithms. Humans contribute to biases in the labour market. A study revealed that resumes including "white names" obtain 50 percent more interview callbacks compared to those with "African-American names" (Bertrand & Mullainathan, 2003, p. 2). This experiment demonstrated

human discrimination in the labour market. Given that artificial intelligence, computer algorithms, and data collection all involve human elements, it is evident how human prejudices can be reflected in computer and AI systems, resulting in discrimination, which can manifest in biased hiring practices, unequal pay, and limited opportunities for certain groups.

An instance of bias in an AI system originates from Amazon.com, Inc.'s 2014 AI hiring initiative. Amazon was compelled to terminate this initiative following the identification of prejudices against women within the employment algorithm. The discrimination originated from the training data, which comprised a decade of submitted resumes to Amazon. Nevertheless, owing to the persistent "male dominance inside the tech industry" (para. 6) resulting from the gender hiring disparity in technology, the algorithm developed a bias towards resumes from men and "penalised resumes that contained the term 'women's'."

The aforementioned example is notable not only for its biased and poor employment algorithm but also for exacerbating the longstanding issue of gender discrimination. This case illustrates how AI decision-making might adversely affect prospective employees and exacerbate broader societal discrimination issues.

Gender discrimination refers to the unfair or unequal treatment of individuals based on their gender. It emerges in various everyday situations and affects opportunities, rights, and social interactions. Females may be discouraged from engaging in scientific or technological studies, whereas males may be inhibited from expressing emotions or choosing caregiving careers. Women may receive inferior compensation compared to men for comparable employment, encounter barriers to promotion, or face enquiries regarding marital and childcare responsibilities during job interviews. Most people expect women to undertake domestic labour and caregiving, whereas they may discourage men from participating equally in household responsibilities. Women may face harassment or restrictions on mobility, while men may experience pressure to conform to ideals of strength and emotional suppression, which can hinder their ability to engage in caregiving and domestic tasks equally. Stereotypical portrayals of men as leaders and women as carers reinforce gender stereotypes and societal expectations.

Gender discrimination in AI occurs when AI systems unfairly evaluate individuals based on their gender. This transpires when algorithms produce unequal outcomes for men, women, or non-binary individuals due to biased data, design choices, or social biases embedded within the system, which can lead to significant disadvantages in areas such as hiring, lending, and law enforcement.

Gender prejudice often originates from historical data that demonstrates unequal societal roles, including male dominance in technical fields and leadership positions. When AI processes such data, it can reinforce stereotypes—associating males with high-paying professions and women with caregiving roles. The formulation of algorithms and the lack of gender diversity in development teams can intensify this bias, leading to discriminatory outcomes in recruitment, credit, healthcare, and advertising, which can perpetuate existing inequalities and limit opportunities for under-represented groups.

This type of prejudice is not attributable to AI itself, as it arises from human sources. In summary, eliminating artificial intelligence from the task would not inherently eradicate or diminish human-induced bias. In conclusion, a biased algorithm may originate from erroneous assumptions of causation based on correlation, extraneous personal factors, skewed or incomplete data sets that exclude specific demographics, or the biases of the individuals who developed the algorithm. All of these characteristics may contribute to bias, which can intensify existing discrimination.

We have discovered that AI decisions may exhibit significant biases that perpetuate societal prejudice. Such biases may arise from conflating causation with correlation, relying on factors unrelated to an algorithm's decision-making, using biased data sets, or incorporating human input, all of which can lead to unfair outcomes in various applications such as hiring, lending, and law enforcement. What methods can be employed to rectify biases in automated decision-making systems?

Eliminate bias.

To eradicate prejudice from AI systems, it is imperative to resolve all aforementioned difficulties. The initial two concerns are intricately connected: presuming causation from two correlated variables introduces bias if the selected variables possess the capacity for discrimination. These variables constitute the "personal factors" that should not influence the algorithm's decision-making process. Nonetheless, the removal of such discriminatory characteristics is more complex than it may seem.

Data supplied to an algorithm may nonetheless embody biased human judgements or mirror historical or societal disparities, even in the absence of sensitive characteristics such as gender, ethnicity, or sexual orientation (Manyika et al., 2019, para. 4). For instance, in the instance of Amazon's gender-biased hiring algorithm, terms like "executed" and "caught" were employed to discriminate against women, as resumes from male candidates frequently included these phrases to a greater extent (Dastin, 2018). Consequently, excluding explicit personal variables

from algorithms is insufficient to avert discrimination. These characteristics may manifest in various aspects of the training data.

A further source of bias arises from distorted data sets. Statistical sets may be distorted due to actual biases or statistics that inadequately represent specific populations. For instance, Amazon's algorithm exhibited bias since the dataset was distorted by existing gender disparities in hiring practices (Dastin, 2018), which led to it favouring male candidates over equally qualified female candidates.

To mitigate prejudice in this instance, the AI program must provide decisions that deviate from prior hiring patterns. When data sets exhibit bias due to an insufficient representation of all demographic groups, they must be rectified. This subject has various perspectives. According to Google's AI department, "Public training data sets will frequently require augmentation to more accurately represent real-world frequencies of individuals" (“Responsible AI Practices", n.d., para. 9). This perspective highlights that data sets may inherently possess bias and require modification and enhancement to mitigate such bias, which can include techniques such as oversampling under-represented groups or incorporating synthetic data to ensure a more balanced representation.

Mitigating the acquisition of morally undesirable biases by learning algorithms necessitates a comprehensive, multi-phase strategy that encompasses the whole AI development process, including problem description, data gathering, model training, and deployment. The fundamental premise is to recognise that bias is usually a manifestation of historic or social disparities ingrained in human data rather than a technical flaw.

Tackling

Consequently, ethical AI development necessitates meticulous data selection, bias assessment, and ongoing oversight to guarantee fairness and accountability.

Auditing datasets

Auditing datasets for representation and fairness entails meticulously examining and assessing training data to guarantee that various social groups are appropriately, proportionately, and equitably represented. This approach aids in detecting imbalances, exclusions, or biased tendencies that may cause AI systems to generate discriminatory or unjust results. Routine dataset audits are crucial for reducing prejudice and guaranteeing that AI systems function ethically, inclusively, and socially responsibly.

Embedding fairness

Incorporating fairness constraints entails intentionally designing algorithms to account for justice and equality in conjunction with accuracy and efficiency in decision-making processes. The system seeks to offer equivalent chances and outcomes for various social groupings (e.g., gender, caste, ethnicity). Conventional algorithms prioritise only accuracy, efficiency, or profitability. When equity is a goal, algorithms must reconcile efficacy with fairness, ensuring that the outcomes do not disproportionately disadvantage any particular group or individual. Moral ideals, like equality, decency, and non-discrimination, are operationalised as quantitative criteria within the algorithm.

Ethical optimisation

Ethical optimisation entails the development of algorithms that enhance not only technical performance (including accuracy, speed, or efficiency) but also align with moral and social values. Ethical values are regarded as limitations or concurrent goals. Principles such as equality, nondiscrimination, transparency, and human dignity are integrated into the optimisation process. Ethical optimisation considers social and cultural aspects, with humans reviewing, validating, and, where required, overriding algorithmic judgements. Human accountability for AI results is unequivocally preserved, hence precluding the transfer of moral responsibility to computers. Crucial in critical domains such as law enforcement, recruitment, healthcare, and welfare allocation

Limiting discriminatory features

Restricting discriminatory characteristics entails prohibiting algorithms from utilising sensitive traits or their hidden alternatives, which may result in unjust or prejudiced outcomes. Characteristics such as race, caste, religion, gender, or handicap are omitted or rigorously regulated during model training. Discrimination transpires when individuals or groups are subjected to unjust or unequal treatment due to inherent or socially ascribed attributes such as race, caste, religion, gender, or handicap. Within the realm of algorithms and artificial intelligence, such prejudice is frequently systemic and deliberate, with potentially severe consequences, including biased decision-making in hiring processes, law enforcement, and access to services. Attributes such as surname, geography, language, or occupation may serve as proxies for race or caste. Discrimination based on caste goes against the constitutional values of equality, dignity, and social justice, making it morally necessary to stop it. An influential model of racial and gender discrimination posits that such discrimination cannot

occur in a perfectly competitive environment, as a company that declines to hire specific workers or serve certain customers would forfeit sales and fail to recruit the most productive employees, ultimately risking bankruptcy. Consequently, numerous economists continue to believe that enhancing market competitiveness serves as a feasible alternative to antidiscrimination legislation. It functions within the model, provided that any pertinent information related to the actual situation, such as the specific barriers faced by marginalised groups, is discarded.

Knock your brain

1The primary reason learning algorithms acquire moral bias is:

A. High computational power

B. Learning from historically biased data

C. Excessive model accuracy

D. Lack of automation

2 Which of the following helps directly detect algorithmic bias?

A. Increasing dataset size
B. Bias and fairness audits
C. Faster processors
D. Encryption

3 Embedding fairness constraints into a model relates to:

A. Data storage
B. Ethical governance
C. Algorithm design
D. User feedback

4 Bias that emerges after deployment is best addressed through:

A. One-time training
B. Ignoring edge cases
C. Continuous monitoring and updates
D. Removing the system

5 Optimizing AI systems only for accuracy may lead to:

A. Fairer outcomes

B. Reduced bias

C. Amplification of existing social biases

D. Perfect neutrality

Unleash your potential

1.What measures may be used to avoid learning algorithms from developing ethically undesirable biases? Engage in discourse.

2.Elucidate the origins of algorithmic bias in artificial intelligence systems.

3.Examine the significance of ethical norms and equity in machine learning.